

Low dimensional counter examples to the Hot Spots conjecture

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1 Introduction

Let $\Omega \subset \mathbb{R}^d$ be a connected, bounded domain with a sufficiently regular boundary $\partial\Omega$. Let $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$ be the eigenvalues of the Neumann Laplacian on Ω such that

$$\begin{aligned} -\Delta\varphi_j &= \lambda_j\varphi_j \text{ in } \Omega, \\ \partial_{\vec{n}}\varphi_j &= 0 \quad \text{in } \partial\Omega. \end{aligned} \tag{1}$$

where $\varphi_0, \varphi_1, \varphi_2, \dots$ denote the corresponding eigenfunctions. Rauch's hot spots conjecture is commonly stated as:

Conjecture 1.1. *For “nice” enough domains Ω , the first eigenfunction φ_1 attains its global extrema at the boundary $\partial\Omega$.*

In a 1975 lecture, J. Rauch [1] introduced the conjecture as a problem in heat conduction: Suppose Ω is some uniformly heat-conducting medium, that is insulated on the boundary. Let $u(t, x)$ be the temperature at $x \in \Omega$ at time t , and suppose the initial temperature distribution is given by u_0 . Rauch posed the question of where the function $u(t, \cdot)$ achieves its maximum as $t \rightarrow \infty$. The answer is given by Conjecture 1.1, which can be stated as follows: *For generic initial conditions u_0 , the point at which $u(t, \cdot)$ achieves its maximum converges to a point on the boundary.*

To explain why this statement is equivalent to Conjecture 1.1 and define generic initial conditions, we note the temperature distribution can be modeled as

$$\begin{aligned} \partial_t u &= \Delta u \text{ in } (0, \infty) \times \Omega, \\ \partial_{\vec{n}} u &= 0 \quad \text{in } (0, \infty) \times \partial\Omega. \end{aligned} \tag{2}$$

By separating the variables t and x , we can rewrite u as

$$u(t, x) = \sum_{j \geq 0} \langle u_0, \varphi_j \rangle \varphi_j(x) e^{-\lambda_j t}. \tag{3}$$

Since φ_0 is constant, this shows that u converges to the mean $\bar{u} = \int_{\Omega} u_0$ as $t \rightarrow \infty$ in the $L^2(\Omega)$ sense. Convergence in the L^∞ sense follows by showing that, under mild regularity conditions on Ω , the functions $\{u(t, \cdot)\}_t$ are uniformly bounded and uniformly equicontinuous. Therefore, assuming for now that $\lambda_2 > \lambda_1$, for large t , the “shape” of $u(t, \cdot)$ will be dominated by the first eigenfunction in the sequence, φ_{j_0} for which $\langle u_0, \varphi_{j_0} \rangle \neq 0$. We say the initial condition u_0 is *generic* if $\langle u_0, \varphi_1 \rangle \neq 0$. Hence, under generic initial conditions, $\tilde{u}(t, x) = e^{\lambda_1 t}(u(t, x) - \bar{u})$ achieves its extrema at the same points as u , and

$$\tilde{u}(t, x) = \langle u_0, \varphi_1 \rangle \varphi_1 + \sum_{j \geq 2} \langle u_0, \varphi_j \rangle \varphi_j(x) e^{-(\lambda_j - \lambda_1)t}. \quad (4)$$

As $t \rightarrow \infty$, the function $\tilde{u}$ must converge exponentially fast to the function $\langle u_0, \varphi_1 \rangle \varphi_1$, and in particular, the extremal points of u converge subsequentially to the extremal points of φ_1 .

Non-generic initial conditions. The requirement that the initial condition be generic is essential. If we naively formulate the hot spots conjecture as stating that the point $x_0 \in \Omega$ at which $u(t, \cdot)$ achieves its maximum tends to the boundary as $t \rightarrow \infty$ for any initial condition, it is straightforward to construct a counterexample. Suppose $\Omega \subset \mathbb{R}^2$ is the unit disk and u_0 is a radially symmetric heat distribution that achieves a strict maximum at the origin. By symmetry, $u(t, \cdot)$ will also achieve its maximum at the origin at any time t . Such an initial condition, however, is never generic. The first non-constant eigenfunction φ_1 is antisymmetric, and therefore $\langle \varphi_1, u_0 \rangle = 0$ (see Figure 1).

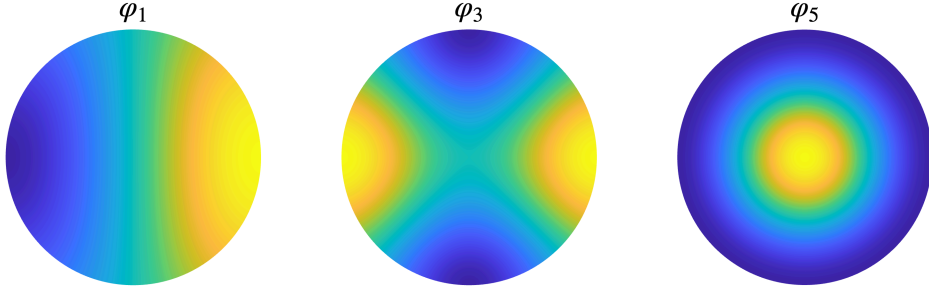


Figure 1: The first eigenfunctions of $-\Delta$ on a disk. Due to rotational symmetry, each eigenspace is two-dimensional.

Multiple ground eigenfunctions The principal eigenspace can be two- or more dimensional; this is often the case if the domain Ω is highly symmetric, such as a square or a disk. In such cases, there are several variations of the conjecture, depending on whether it should hold for all or only some eigenfunctions in the eigenspace. Following Bañuelos and Burdzy [2], we distinguish three formulations:

(HS1) For *every* eigenfunction φ_1 corresponding to λ_1 which is not identically 0, and all $y \in \Omega$, we have $\inf_{x \in \partial\Omega} \varphi_1(x) < \varphi_1(y) < \sup_{x \in \partial\Omega} \varphi_1(x)$.

(HS2) For *every* eigenfunction φ_1 corresponding to λ_1 and all $y \in \Omega$, we have $\inf_{x \in \partial\Omega} \varphi_1(x) \leq \varphi_1(y) \leq \sup_{x \in \partial\Omega} \varphi_1(x)$.

(HS3) There *exists* an eigenfunction φ_1 corresponding to λ_1 which is not identically 0, and such that for all $y \in \Omega$, we have $\inf_{x \in \partial\Omega} \varphi_1(x) < \varphi_1(y) < \sup_{x \in \partial\Omega} \varphi_1(x)$.

In this work, we will build counterexamples with one-dimensional eigenspaces and eigenfunctions which attain a strict maximum in the interior, therefore disproving all three versions simultaneously.

What are nice enough domains? The open question is identifying the classes of domains for which the conjecture holds and those for which it fails. The difficulty is that the principal eigenfunction of the Laplacian can be computed in closed form only for very few domains. For example, on a rectangle φ_1 is a product of trigonometric functions and the statement of the conjecture is easily verified.

Kawohl [3, Corollary 2.15] showed that (HS1) holds for cylindrical domains $\Omega = \Omega_0 \times (0, l) \subset \mathbb{R}^d$ where Ω_0 is bounded with $\partial\Omega_0$ of class $C^{0,1}$. Furthermore, Kawohl conjectured that the property holds for all convex domains [3, p.56]. Bañuelos and Burdzy [2] later proved that (HS1) holds for certain triangles and for long convex planar domains with sufficient symmetry. Conversely, Burdzy

and Werner [4] constructed a counterexample consisting of a bounded planar domain with two holes, the number of holes was later reduced to one [5]. For numerical counterexamples, see [6].

The conjecture was believed to be true for convex sets until it was recently disproven [7] in high dimensions. This recent work is the subject of the next subsection.

1.1 Convex sets can have interior hot spots

The counterexample in [7] is built using a sequence of convex sets $\Omega_d \subset \mathbb{R}^{d+2}$ which are of the following nature:

$$\Omega_d = \left\{ (x, w) \in Q \times \mathbb{R}^{d+1} : |w| \leq \frac{1}{2} \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}} \right) \right\}, \quad (5)$$

where $Q \subset \mathbb{R}^2$ is a rectangle and $V : Q \rightarrow \mathbb{R}$ is a convex potential. We now study the sequence of principal eigenfunctions $\varphi_{1,d}$ of Ω_d . First of all, notice that for d large enough $\varphi_{1,d}$ is radial in the second argument w . We can therefore, overloading the notation, write $\varphi_{1,d}(x, r) := \varphi_{1,d}(x, 2d^{-\frac{1}{2}}|w|)$. Under the change of variables $r = 2d^{-\frac{1}{2}}|w|$, the set Ω_d becomes effectively three dimensional

$$\Omega_d = \left\{ (x, r) \in Q \times \mathbb{R}_{\geq 0} : |w| \leq 1 - \frac{V(x)}{d} \right\}, \quad (6)$$

and the eigenvalue equation $-\Delta\varphi_{1,d} = \lambda_{1,d}\varphi_{1,d}$ transforms into

$$\partial_r \varphi_{1,d} = \frac{r}{4} (\Delta_x - \lambda_{1,d}) \varphi_{1,d} + O(d^{-1}). \quad (7)$$

The sequence $\varphi_{1,d}$ converge uniformly to a function $\varphi_{1,\infty} : Q \times [0, 1]$ which satisfies

$$\begin{aligned} \partial_r \varphi_{1,d} &= \frac{r}{4} (\Delta_x - \lambda) \varphi_{1,d} \\ L\varphi_{1,d}(x, 1) &= \lambda_{1,\infty} \varphi_{1,d} \\ \partial_r \varphi_{1,d}(x, r) &= 0 \text{ at } r = 0 \\ \partial_{\bar{n}} \varphi_{1,d} &= 0, \end{aligned} \quad (8)$$

where $L := -\Delta + \nabla V \cdot \nabla$, see [7, Theorem 2.6]. The counterexample is now effectively built at $d = \infty$, i.e. it is shown that there exists a convex potential V such that $\varphi_{1,\infty}$ does achieve its maximum in $Q^\circ \times [0, 1)$. As a result there must be a critical dimension d_{HS} such that for all $d \geq d_{\text{HS}}$ the functions $\varphi_{1,d}$ achieve their maximum in $Q^\circ \times [0, 1)$ as well.

The dimension d_{HS} , at which the hot spot property starts to fail, is not explicit and the naive tracking of constants does not yield a satisfactory bound. In this work we aim to give an explicit lower bound of the dimension for which (HS1-HS3) starts to fail. We do this by solving for the principal eigenfunction φ_1 numerically and then certifying the counterexample using interval arithmetic.

1.2 The method of particular solutions and certified numerics

We draw inspiration from [8], where the authors construct a numerical counterexample to Payne's nodal line conjecture using the method of particular solutions and rigorously certify the counterexample. The idea behind the method is simple. We explain it in context of our eigenvalue problem, but it transfers to any boundary value problem: Choose a set of functions $\{f_j\}_j$ which satisfy the eigenvalue equation $-\Delta f_j = \lambda_* f_j$ in the interior, disregarding the boundary conditions. Any linear combination $\varphi_* = \sum_j c_j f_j$ of functions in $\{f_j\}_j$ also satisfies the eigenvalue equation

$-\Delta\varphi_* = \lambda_*$. In order to obtain an approximation of the true eigenfunction φ_1 we optimize the coefficients $\{c_j\}$ and the approximate eigenvalue λ^* with the objective of minimizing the error on the boundary.

Once we obtained a candidate approximation φ_* , we prove that if the error on the boundary is small $\|\partial_{\vec{n}}\varphi_*\|_{L^\infty(\partial\Omega)}$, then φ_* must be close to the true eigenfunction φ_1 point wise. One limitation of the method of particular solutions is that it does not guarantee the position in the spectrum of the approximation φ_* , that is, φ_* could have small boundary error but be an approximation of the second eigenfunction φ_2 instead of φ_1 . This can be particularly problematic in settings as in [8], where the eigenvalue of interest lies in a cluster. Luckily, in our case we have a large enough spectral gap $\lambda_2 - \lambda_1 > ?$. Furthermore we can show that φ_1 is the only eigenfunction satisfying certain geometric properties which are easy to check for the approximation φ_* .

Since φ_* and $\|\partial_{\vec{n}}\varphi_*\|_{L^\infty(\partial\Omega)}$ are evaluated numerically, they are subject to floating point arithmetic errors. To circumvent this issue, once we have constructed a counterexample φ_* , we compute rigorous enclosures of φ_* and $\|\partial_{\vec{n}}\varphi_*\|_{L^\infty(\partial\Omega)}$ using interval arithmetic.

1.3 Outline of the thesis

This thesis is structured as follows: In Section 2 we introduce the class of sets used in [7] to construct a counterexample. We show that on this class of sets the principal eigenfunction φ_1 is highly symmetrical along some dimensions allowing us to reduce the effective dimension of the set giving the counterexample to three. On this effectively three dimensional set we construct a basis for the method of particular solutions.

In Section 3 we explain the counterexample in [7] and derive an adaptation which can be explicitly computed. In Section 4 we present our numerical implementation of our method of particular solutions and compute the rigorous error bounds. Finally, in Section 5 we prove that with the error bound found in Section 4, there must be a counterexample in finite dimensions.

2 Barrel sets

In this section we introduce the class of *barrel sets* which are used in [7] to construct a counterexample and on which we will base our counterexample as well.

Definition 2.1. Let $Q := [-l_1/2, l_1/2] \times [-l_2/2, l_2/2]$ and let $V : Q \rightarrow \mathbb{R}$ be a convex potential. We define the convex domain

$$F_d(Q, V) := \left\{ (x, w) \in Q \times \mathbb{R}^{d+1} : |w| \leq \frac{1}{2} \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}} \right) \right\}, \quad (9)$$

which we call a barrel domain.

The sets barrel sets get their name from the fact that for an interval I the set $F_1(I, V)$ looks like a barrel, see Figure 2.

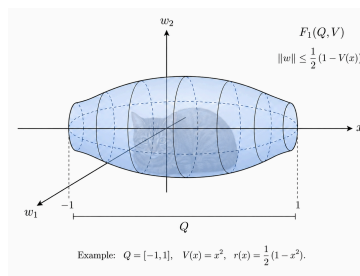


Figure 2: (Preliminary AI slop) The sets F_1 look like barells for $Q = [-1, 1]$

Since the boundary of $F_d(Q, V)$ is not differentiable at points (x, w) where $w \in \partial\Omega$, we define the Neumann eigenfunctions using the weak formulation. A non-zero function $u \in H^1(\Omega)$ is a Neumann eigenfunction with eigenvalue $\lambda \geq 0$ if it satisfies the equality $\int_{\Omega} \nabla u \cdot \nabla v \, dx = \lambda \int_{\Omega} uv \, dx$, for all $v \in H^1(\Omega)$.

Symmetries of the potential V translate to symmetries of the eigenfunctions, this is the subject of the following lemmas.

Lemma 2.2. *Any Neumann eigenfunction, that is not radial in the w -coordinate, has eigenvalue at least 4.*

Proof. In cylindrical coordinated the Laplacian is $\Delta = \Delta_x + \partial_r^2 + \frac{d}{r} \partial_r + \frac{1}{r^2} \Delta_{S_d}$. We separate $\varphi_{1,d}$ into eigenfunctions of $-(\Delta_x + \partial_r^2 + \frac{d}{r} \partial_r)$ and $-\frac{1}{r^2} \Delta_{S_d}$, that is

$$\varphi_{1,d}(x, |w|) = \sum_{\ell \geq 0, m} A_{\ell,m}(x, |w|) Y_{\ell,m}(\theta), \quad (10)$$

where $-\Delta A_{\ell,m} = \lambda_{\ell,m} A_{\ell,m}$ and $-\Delta Y_{\ell,m} = \ell(\ell + d - 1) Y_{\ell,m}$. Each component $A_{\ell,m} Y_{\ell,m}$ is itself an eigenfunction of $-\Delta$ corresponding to the eigenvalue $\lambda_{1,d}$. Therefore, each component minimizes the Rayleigh quotient and thus

$$\lambda_{1,d} = \frac{\int |\nabla A_{\ell,m} Y_{\ell,m}|^2}{\int |A_{\ell,m} Y_{\ell,m}|^2} \geq \ell(\ell + d - 1) \frac{\int_{\Omega} \int_0^{\rho(x)} |A_{\ell,m}|^2 r^{d-2} \, dr \, dx}{\int_{\Omega} \int_0^{\rho(x)} |A_{\ell,m}|^2 r^d \, dr \, dx} \geq \frac{\ell(\ell + d - 1)}{\rho_{\max}^2}. \quad (11)$$

Therefore, whenever $A_{\ell,m} \neq 0$ for some $\ell \geq 1$, we have $\lambda_1 \geq \frac{d}{\rho_{\max}^2}$ and $\rho_{\max} = \frac{1}{2} \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}} \right) \leq \frac{5}{8} \sqrt{d}$

Here I think we are giving away too much: $\frac{1}{2} \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}} \right) \leq \frac{1}{2} \left(\sqrt{d} - \frac{V(0)}{\sqrt{d}} \right) \approx \frac{1}{2} \sqrt{d}$. Then we can conclude that if the eigenfunction is not radial then, $\lambda_{1,d} \geq 4$.

, so $\lambda_{1,d} \geq \frac{64}{25} = 2.56$ whenever $A_{\ell,m} \neq 0$ for some $\ell \geq 1$. Since $\lambda_{1,d} \leq 2$ for d large enough, only $A_{0,0} Y_{0,0}$ survives which is constant in w . $\square$

Lemma 2.3.

$$\lambda_1 < 3.9 \quad (12)$$

Proof. Use the numerical counterexample in some way. Worst case, compute its Rayleigh quotient. $\square$

Corollary 2.3.1. *The principal eigenfunction $\varphi_{1,d}$ on $F_d(V)$ is radial in w for $d \geq 4\|V\|_{\infty}$*

How the above scales with scaling of the domain. If we rescale Q to sQ and V to $V_s = V(x/s, y/s)$, then the lower bound if the eigenfunction is not radial stays the same. However, the eigenvalue $\lambda_{1,d}$ multiplies by $\frac{1}{s^2}$, hence the conclusion of the above lemma holds for sQ, V_s whenever $\lambda_{Q,V} < 2.5s^2$.

Lemma 2.4. *Assume that $V(-x_1, x_2) = V(x_1, x_2) = V(x_1, -x_2)$. Then every Neumann eigenspace of $F_d(Q, V)$ has an orthonormal basis whose elements are even/odd, odd/even, even/even or odd/odd in (x_1, x_2) .*

Proof. Let $\Omega = F_d(Q, V)$ and define the reflections

$$r_1(x_1, x_2, w) = (-x_1, x_2, w), \quad r_2(x_1, x_2, w) = (x_1, -x_2, w). \quad (13)$$

Since Q and V are invariant under both reflections, $r_i(\Omega) = \Omega$ for $i = 1, 2$. Hence the operators $R_i u := u \circ r_i$ are unitary involutions on $L^2(\Omega)$ and preserve $H^1(\Omega)$. If u is an eigenfunction with eigenvalue λ , then for every test function v ,

$$\int_{\Omega} \nabla(R_i u) \cdot \nabla v \, dz = \int_{\Omega} \nabla u \cdot \nabla(R_i v) \, dz = \lambda \int_{\Omega} u R_i v \, dz = \lambda \int_{\Omega} (R_i u) v \, dz. \quad (14)$$

Hence $R_i u$ is an eigenfunction with the same eigenvalue, so R_i leaves each eigenspace invariant.

The two reflections commute. Therefore their restrictions to any eigenspace E_{λ} are commuting self-adjoint operators on the finite-dimensional Hilbert space E_{λ} , and can be simultaneously diagonalized. For a common eigenvector u we have $R_i u = \varepsilon_i u$ with $\varepsilon_i \in \{+1, -1\}$. The sign $+1$ means that u is even in x_i , while the sign -1 means that u is odd in x_i . This gives the claimed parity basis. In particular, every eigenfunction in a simple eigenspace has one of these four parity types. $\square$

Lemma 2.5. *Assume that $V(-x_1, x_2) = V(x_1, x_2) = V(x_1, -x_2)$ and let λ_1 be the first non-zero Neumann eigenvalue of $F_{d(Q,V)}$. Then the eigenspace for λ_1 does not contain odd/odd functions.*

Proof. By Courant's nodal domain theorem, every eigenfunction for λ_1 has at most two nodal domains. $\square$

Lemma 2.6. *The lowest radial eigenvalue in the odd/even sector is strictly less than the lowest radial eigenvalues of each of the other sectors.*

Proof. Verify numerically. $\square$

Corollary 2.6.1. *The first eigenspace is simple and the first eigenfunction is odd/even.*

We will also need a bound on the spectral gap. If we want to compute the second eigenfunction with assumed symmetry, we need to show that the lowest eigenvalue in the corresponding sector is less than the eigenvalues in the remaining sectors. Another way would be to still compute the coefficients assuming a parity, and then setting the coefficients for eigenfunctions in other sectors to zero at verification time.

As a result of Lemma 2.2 and Corollary 2.6.1, after the change of variables $r = 2|w|/\sqrt{d}$, we can restrict the eigenvalue problem to the space

$$\Omega_d(V) = \left\{ (x, w) \in [0, l_1/2] \times [0, l_2/2] \times \mathbb{R}_{\geq 0} : r \leq 1 - \frac{V(x)}{d} \right\} \quad (15)$$

with Dirichlet boundary conditions at the face $x_1 = 0$ and Neumann boundary conditions everywhere else.

2.1 MPS basis for Barrel sets

We aim to construct a set of functions $\{X_{j,k} R_{j,k}\}_{j,k}$ such that any linear combination of those functions

$$\varphi_*(x, r) = \sum_{j,k} c_{j,k} X_{j,k}(x) R_{j,k}(r), \quad (16)$$

satisfies $-\Delta\varphi_* = \lambda_*\varphi_*$ in F_d . Under the change of variables $r = 2|w|/\sqrt{d}$ the Laplacian transforms into

$$\Delta = \Delta_x + \frac{4}{d}\partial_r^2 + \frac{4}{r}\partial_r. \quad (17)$$

We choose $X_{j,k}$ and $R_{j,k}$ such that

$$-\Delta_x X_{j,k} = \lambda_{x,j,k} X_{j,k}, \quad (d^{-1}\partial_r^2 + r^{-1}\partial_r)R_{j,k} = \lambda_{r,j,k} R_{j,k} \quad (18)$$

for real numbers $\lambda_{x,j,k}$ and $\lambda_{r,j,k}$ satisfying $\lambda_{x,j,k} + \lambda_{r,j,k} = \lambda_*$. For Δ_x the eigenfunctions are straight forward:

$$X_{j,k}(x) = \sin\left(x_1 \frac{(2j+1)\pi}{l_1}\right) \cos\left(x_2 \frac{2k\pi}{l_2}\right) \quad (19)$$

this prescribes $\lambda_{x,j,k} = \left(\frac{(2j+1)\pi}{l_1}\right)^2 + \left(\frac{2k\pi}{l_2}\right)^2$ and $\lambda_{r,j,k} = \lambda_* - \lambda_{x,j,k}$. Observe that $\lambda_{r,j,k}$ can be negative, in fact it is negative for most j, k . As a result $R_{j,k}$ has two branches

$$R_{j,k}(r) \sim \begin{cases} r^{-\alpha_d} I_\alpha(\beta_{d,j,k} r) & \text{if } \lambda_{r,j,k} < 0 \\ r^{-\alpha_d} J_\alpha(\beta_{d,j,k} r) & \text{if } \lambda_{r,j,k} \geq 0 \end{cases} \quad (20)$$

where $\alpha_d = \frac{d-1}{2}$ and $\beta_{d,j,k} = \sqrt{\frac{d}{4}|\lambda_{r,j,k}|}$. See Appendix A.1 for a proof of $-\Delta X_{j,k} = \lambda_{x,j,k} X_{j,k}$.

3 Constructing a counterexample set

In this section we aim to explain the counterexample from [7] and how we adapted it to a version which can be computed explicitly. Since our proof relies on verified numerics, we do not proof any properties of the counterexample set rigorously, but rather explain the intuition behind the reason why we expect the counterexample to work.

The counterexample is effectively built in the limit $d \rightarrow \infty$, we first have to understand how to eigenvalue problem behaves in this limit. This is the subject of the next subsection.

3.1 The effective problem in the limit of dimension

The sequence of principal eigenfunctions $\varphi_{1,d}$ of F_d at dimension d converges to a function h that is an eigenfunction of $L = -\Delta + \nabla V \cdot \nabla$ at $r = 1$ and satisfies the following initial value problem:

$$\begin{aligned} \partial_r h &= -\frac{r}{4}(\Delta_x + \lambda)h \text{ for } x \in Q, r \in [0, 1) \\ h(x, 1) &= \psi_1(x) \text{ for } x \in Q \\ \partial_{\vec{n}} h &= 0 \text{ for } x \in \partial Q, \end{aligned} \quad (21)$$

here ψ_1 is the first non-trivial eigenfunction of L . The first equation follows immediately from the eigenvalue equation $-\Delta\varphi_{1,d} = \lambda_{1,d}\varphi$ since

$$\Delta = \Delta_x + \frac{4}{d}\partial_r^2 + \frac{4}{r}\partial_r \rightarrow \Delta_x + \frac{4}{r}\partial_r. \quad (22)$$

For the boundary condition we examine rayleigh quotient

$$\frac{\int_Q \int_0^{\rho_d(V)} |\nabla u|^2 r^d dr dx}{\int_Q \int_0^{\rho_d(V)} |u|^2 r^d dr dx} = \frac{\int_Q |\nabla u|^2 \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}}\right)^{d+1} dx}{\int_Q |u|^2 \left(\sqrt{d} - \frac{V(x)}{\sqrt{d}}\right)^{d+1} dx} \rightarrow \frac{\int_Q |\nabla u|^2 e^{-V(x)} dx}{\int_Q |u|^2 e^{-V(x)} dx}. \quad (23)$$

This is the Rayleigh quotient for the Neumann problem $Lu = \lambda u$ on Q .

The boundary value problem in Equation 21 can be transformed into a reaction-diffusion initial value problem by the change of variables $t = \frac{1-r^2}{8}$, we obtain

$$\begin{aligned} \partial_t h &= \Delta_x h + \lambda h \text{ for } t \in (0, 1/8] \\ h(x, 0) &= \psi_1(x) \\ \partial_n h &= 0 \text{ on the spatial boundary.} \end{aligned} \tag{24}$$

The goal is now to find a suitable initial value ψ_1 .

3.2 The counterexample in [7] and adapting it for computation

In this section we explain the main idea behind the counterexample in [7] and how to build the corresponding potential explicitly in a way suitable for computation. We begin our discussion with the initial condition for Equation 24 we are aiming for. This is perhaps best explained by a plot of ψ_1 , see Figure 3.



Figure 3: Left: The eigenfunction on the full domain. Center: The green region magnified. Right: Surface plot of the region.

The axis of the Barrel decomposes into two parts: The “core” $A_{\text{core}} = [0, \pi/2] \times [0, 1]$ and the “wing” $A_{\text{wing}} = [\pi/2, \pi/2 + m]$. On a macroscopic scale we want ψ_1 to look like $\sin(x_1)$ in A_{core} , that is, like the principal eigenfunction of $-\Delta$ on A_{core} . In A_{wing} we want ψ_1 to be a constant extension of the core function. On a microscopic scale we want ψ_1 to be perturbed in such a way that at the interface of A_{core} and A_{wing} we have $\psi_1(x) \approx q(x_2) + \text{const.}$, where q is as in Figure 4. This profile should extend into A_{wing} but lose its high points at $x_2 = \pm 1.0$ before the end of the wing, see Figure 4. The initial datum ψ_1 achieves its maximum at $x = (\pi/2 + m, 0)$, which is on the boundary. The heat extensions of the profile q and its trimmed version both achieve their maximum at the origin at any time. However, the heat extension of q will be strictly larger than the one of the trimmed versions, due to the additional energy in the tails. The full solution h will capture the same phenomenon, but due to the additional reaction term in Equation 24, the extension h will be strictly larger near $t = \frac{1}{8}$ (or $r = 0$),

$$h(x, r) = \sum_{j,k} c_{j,k} X_{j,k}(x) \exp\left(-r^2 \frac{\lambda_{x,j,k} - \lambda_1}{8}\right), \tag{25}$$

hence the maximum of h will be in the interior near $x_2 = 0, r = 0$ and $x_1 \in (\frac{\pi}{2}, \frac{\pi}{2} + m)$.

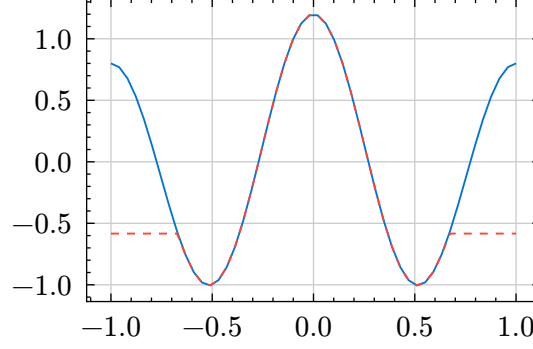


Figure 4: Cross section of ψ_1 at the core-wing interface and at the end of the wing.

3.2.1 Deriving a potential to obtain the desired initial datum in the core

In order to build a potential V such that ψ_1 has the before-mentioned properties in A_{core} , we consider the operator $L = -\Delta + \nabla V \cdot \nabla$ to be a perturbation of the Laplacian $-\Delta$. Let $V_\varepsilon = \varepsilon^{-1}V$, then we can write L as

$$L = -\Delta + \varepsilon \nabla V_\varepsilon \cdot \nabla. \quad (26)$$

Expand both λ_1 and ψ_1 in powers of ε ,

$$\begin{aligned} \lambda_1 &= \tilde{\lambda}_1 + \varepsilon \mu + O(\varepsilon^2) \\ \psi_1 &= \tilde{\psi}_1 + \varepsilon \beta + O(\varepsilon^2). \end{aligned} \quad (27)$$

Substituting the expansions into $L\psi_1 = \lambda_1\psi_1$ and collecting terms of the same order in ε we get

$$\begin{aligned} \tilde{\psi}_1 &= \sqrt{\frac{\pi}{2}} \sin(x_1), \quad \tilde{\lambda}_1 = 1 \\ (-\Delta - 1)\beta &= \sqrt{\frac{\pi}{2}} (\mu \sin(x_1) - (\partial_1 V_\varepsilon) \cos(x_1)). \end{aligned} \quad (28)$$

We already obtain the correct macroscopic behavior of ψ_1 since ε is small. The rest of the desired properties will be enabled by β . Let β_0 be

- antisymmetric in x_1 ,
- symmetric in x_2 ,
- have Neumann boundary data with $\beta_0(\pi/2, x_2) = q(x_2) + \text{const.}$, and
- satisfy $(\Delta + 1)\beta_0(\pi/2, x_2) = 0$.

The function

$$V_0(x_1, x_2) = -\sqrt{\frac{2}{\pi}} \int_0^{x_1} \frac{(\Delta + 1)\beta_0(s, x_2)}{\cos(s)} ds \quad (29)$$

satisfies

$$(-\Delta - 1)\beta_0 = -\sqrt{\frac{\pi}{2}} (\partial_1 V_0) \cos(x_1). \quad (30)$$

From the function V_0 we obtain the convex potential $V(x) = V_0(x) + \frac{1}{2}M\|x\|^2$, for M large enough. In practice we get a lower bound for M by computing the least eigenvalue of the hessian of V . Let $s(x_1), \mu$ be a solution of the ODE $-\partial_1^2 s(x_1) - s(x_1) = \sqrt{\frac{\pi}{2}} (\frac{\mu}{M} \sin(x_1) - x_1 \cos(x_1))$ on $[-\pi/2, \pi/2]$ with Neumann boundary conditions. Then, $\beta(x_1, x_2) = \beta_0(x_1, x_2) + Ms(x_1)$ satisfies

$$\begin{aligned}
(-\Delta-1)\beta(x_1, x_2) &= \sqrt{\frac{\pi}{2}} \left(-(\partial_1 V_0) \cos(x_1) + M \left(\frac{\mu}{M} \sin(x_1) - x_1 \cos(x_1) \right) \right) \\
&= \sqrt{\frac{\pi}{2}} (\mu \sin(x_1) - (\partial_1 V) \cos(x_1)),
\end{aligned} \tag{31}$$

as desired.

Constructing β_0 We use the ansatz

$$\beta_0(x_1, x_2) = f_0(x_1) + \sum_{n=1}^N q_n f_n(x_1) \cos(n\pi x_2) \tag{32}$$

where $f_n(x_1) = \sum_{j=0}^{J-1} a_{n,j} T_{2j+1}(2x_1/\pi)$. The symmetry constraints on β are automatically satisfied. By imposing the constraint $\sum_j a_{n,j} = 1$ for all $n \geq 1$ we obtain

$$\begin{aligned}
\beta_0\left(\frac{\pi}{2}, x_2\right) - f_0\left(\frac{\pi}{2}\right) &= \sum_{n=1}^N q_n \left(\sum_j a_{n,j} T_{2j+1}(1) \right) \cos(n\pi x_2) \\
&= \sum_n q_n \cos(n\pi x_2) = q(x_2).
\end{aligned} \tag{33}$$

From the Neumann boundary condition $f'_n(\frac{\pi}{2}) = 0$, we get the constraint

$$\sum_j a_{n,j} (2j+1)^2 = n^2 \pi^2 - 1. \tag{34}$$

Finally, $(\Delta+1)\beta_0(\pi/2, x_2) = 0$ is imposed by

$$f''_n\left(\frac{\pi}{2}\right) = (n^2 \pi^2 - 1) f_n\left(\frac{\pi}{2}\right) \Rightarrow \sum_j a_{n,j} (2j+1)^2 ((2j+1)^2 - 1) = (n^2 \pi^2 - 1) \sum_j a_{n,j}. \tag{35}$$

It is possible to satisfy the three constraints with $J = 3$. However, we choose $J > 3$, in order to have some degrees of freedom to minimize the convexification cost M . The zeroth mode f_0 has the same constraints except $f_0(\frac{\pi}{2}) = 1$, but instead $\langle f_0, \sin(x_1) \rangle = 0$.

3.2.2 Extending to the wings

In the previous paragraph we built V_0 such that the *Neumann* eigenfunction ψ_1 has the prescribed boundary profile q . In order to maintain this profile at the core-wing interface while extending the domain, we need V to act as a virtual Neumann boundary condition.

The probabilistic interpretation of the semigroup. Let $P_t = e^{tL}$ be the semigroup corresponding to L . We have that $P_t \psi_1 = e^{-\lambda_1 t} \psi_1$. Let X_t denote the stochastic process described by $dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t$, then the action of P_t on a function f can be written as

$$P_t f(x) = \mathbb{E}[f(X_t) \mid X_0 = x]. \tag{36}$$

The Neumann boundary condition is encoded by having dX_t reflect at the boundary. Therefore, by letting ∇V increase sharply at the interface between A_{core} and A_{wing} , we can enforce a virtual Neumann boundary condition at ∂A_{core} . We can achieve this by setting $V(x) = V(\frac{\pi}{2}, 1.0) + 10^7(|x_1| - \frac{\pi}{2})$ in A_{wing} , this way particles coming from A_{core} have a high probability of reflecting near $x_1 = \frac{\pi}{2}$, this simulates the reflecting boundary conditions.

Next, we establish that the eigenfunction ψ_1 is approximately constant along the flow lines of ∇V in A_{wing} . Indeed, let h be the extension of ψ_1 , then, near $r = 1$ we have $\nabla_x V \cdot \nabla_x h(\cdot, r) \approx \Delta_x h +$

$\lambda_1 h = 0$. Therefore, since h is smooth, we have $\nabla V \cdot \nabla \psi_1 \approx 0$. As a result, we can transport the profile at the core-wing interface via the flow lines of ∇V .

4 Solving for the eigenfunction Numerically

In this section we explain how we compute approximations of the Neumann eigenfunctions of the Barrel sets. We use the Method of Particular Solutions (MPS) as described by [9]. That is, we write our approximate eigenfunction ϕ_* as a finite linear combination $\sum_j c_j \phi_j$ of the basis functions derived in Section 2.1. The basis functions are such that $-\Delta \phi_j = \lambda_*$ for all j any chosen λ_* which is a parameter of ϕ_j . As a result, the approximation ϕ_* satisfies $-\Delta \phi_* = \lambda_*$ exactly. We aim to find coefficients c_j and a value λ_* , which minimize the error in the Neumann boundary condition. We sample m_B points (x_i, r_i) from the boundary $\Gamma_2 \subset \partial\Omega$. Using these collocation points we define the matrix

$$A_B(\lambda) = [\partial_{\vec{n}} \phi_j(x_i, r_i)]_{i,j}. \quad (37)$$

Finding coefficients $c \in \mathbb{R}^N$ that minimize $\|A_B c\|^2 = \sum_i |\partial_{\vec{n}} \phi_*(x_i, r_i)|^2$ under the constraint $\|c\| = 1$ is equivalent to finding the smallest singular value of A_B and its corresponding right singular vector. The problem is, that the condition number of A_B grows exponentially with the number of basis modes N , and, for large N , there exist linear combinations of ϕ_j that approximate the zero function in the interior [9]. A solution for this issue was proposed by [9], and also used by [8]: We sample m_I points from the interior of Ω to construct the matrix $A_I(\lambda) = [\phi_j(x_i, r_i)]_{i,j}$. We factorize

$$A = \begin{pmatrix} A_B \\ A_I \end{pmatrix} = \begin{pmatrix} Q_B \\ Q_I \end{pmatrix} R = QR, \quad (38)$$

to obtain an orthonormal basis Q_B for the boundary. We then solve the problem

$$c = \arg \min_{\|c\|=1} \|Q_B c\| \quad (39)$$

by computing the right singular vector of Q_B .

4.1 Guaranteed error bounds using interval arithmetic

Here goes the theroem that we have an approximation φ_* at dimension d with a certain boundary error.

5 Certifying the numerical counterexample

In this section we aim to show that whenever we have an approximate eigenfunction φ_* of φ_1 obtained through the method of particular solutions, which has small error on the boundary conditions, then φ_* is close to φ pointwise. The theorems used to certify this all have in common that they need some type of a priori bounds of the eigenvalues.

5.1 Bounds of the eigenvalue

The problem with Moler-Payne and similar methods is that they do not give the position in the spectrum. We first separate the first two eigenvalues using a Galerkin method and certify the bounds as described in [10].

Theorem 5.1.1 (Theorem 3.2 in [10]). *Suppose that for the interpolation error Π_h it holds that $\int |\nabla(u - \Pi_h u)|^2 dx \leq C_h \int |u - \Pi_h u|^2 dx$ for all $u \in H^1$. Then*

$$\frac{\lambda_{k,h}}{1 + \lambda_{k,h} C_h^2} \leq \lambda_k. \quad (40)$$

5.1.1 Bounding the global eigenvalues by the eigenvalues in the core

In this section we prove that we can approximate the spectrum of L on Q by the spectrum of L restricted to a subset of Q which carries most of the mass.

Let $d\mu = \frac{1}{Z}e^{-V}$, assume $\|\varphi_1\|_{L^2(\mu)} = 1$, then

$$\begin{aligned} \lambda_1 &= \int_Q |\nabla \varphi_1|^2 d\mu \\ &\geq \inf_f \frac{\int_{\text{Core}} |\nabla f|^2 d\mu}{\int_{\text{Core}} |f|^2 d\mu} \int_{\text{Core}} |\varphi_1|^2 d\mu \\ &= \lambda_1^{\text{Core}} \int_{\text{Core}} |\varphi_1|^2 d\mu. \end{aligned} \quad (41)$$

We have

$$\int_{\text{Core}} |\varphi_1|^2 d\mu = 1 - \int_{\text{Wing}} |\varphi_1|^2 d\mu \quad (42)$$

The simplest bound of this is

$$\int_{\text{Wing}} |\varphi_1|^2 d\mu \leq \|\varphi_1\|_\infty^2 \mu(\text{Wing}). \quad (43)$$

Now we bound the infinity norm using Wang-Li-Yau: $\|\varphi_1\|_\infty \leq e^{\lambda_1 t} C \|\varphi_1\|_2 = e^{\lambda_1 t} C$.

Now let's try to lower bound λ_2 . Let $S = \text{span}\{\varphi_1, \varphi_2\}$ (first two nonconstant eigenfunctions)

$$\begin{aligned} \lambda_2^{\text{Core}} &\leq \max_{1 \perp h \in S} \frac{\int_{\text{Core}} |\nabla h|^2 d\mu}{\int_{\text{Core}} |h|^2 d\mu} \\ &\leq \max_{1 \perp h \in S} \frac{\int_Q |\nabla h|^2 d\mu}{\int_Q |h|^2 d\mu} \frac{\int_Q |h|^2 d\mu}{\int_{\text{Core}} |h|^2 d\mu} \\ &\leq \lambda_2 \max_{1 \perp h \in S} \frac{\int_Q |h|^2 d\mu}{\int_{\text{Core}} |h|^2 d\mu} \end{aligned} \quad (44)$$

we have $\int_Q |\varphi_1 + \varphi_2|^2 d\mu =$

5.2 Pointwise bounds of the eigenfunction

Let $\partial\Omega = \Gamma_0 \cup \Gamma_1 \cup \Gamma_2$ where $\Gamma_0 = \{(x, r) \in \partial\Omega : x_1 = 0\}$, $\Gamma_1 = \{(x, r) \in \partial\Omega : |x_1| = 1 \vee |x_2| = 1 \vee r = 0\}$ and $\Gamma_2 = \{(x, r) \in \partial\Omega : r = 1 - \frac{V(x)}{d}\}$. We construct φ_* such that the boundary conditions on $\Gamma_0 \cup \Gamma_1$ are satisfied exactly.

5.2.1 Pointwise bounds of the eigenfunction in $d = \infty$ using Wang-Li-Yau

Let φ_* be an approximate eigenfunction, i.e. $\|(L - \lambda_*)\varphi_*\|_\infty \leq \varepsilon$ and $\|\varphi_*\|_2 = 1$.

Denote by P_t the semigroup for L , such that $\partial_t P_t = -LP_t$. Let $Q_t := e^{\lambda_1 t} P_t$, then $Q_t \varphi_1 = \varphi_1$ and $Q_t \varphi_* \rightarrow \varphi_1$ as $t \rightarrow \infty$. Therefore, we decompose the approximation error into

$$\|\varphi_* - \varphi_1\|_\infty \leq \|Q_t \varphi_* - \varphi_*\|_\infty + \|Q_t \varphi_* - \varphi_1\|_\infty. \quad (45)$$

The first term we can bound by $\int_0^t \|Q_s(L - \lambda_1)\varphi_*\|_\infty ds$ since $\partial_t Q_t = -(L - \lambda_1)Q_t$. The second term we can bound by $C_t \|\varphi_* - \varphi_1\|_2$ after applying Wang-Li-Yau. Or even better, if we can send $t \rightarrow \infty$, then the second term vanishes completely.

To bound the first term, we could use the naive bound $\|Q_s(L - \lambda_1)\varphi_*\|_\infty \leq e^{\lambda_1 s} \|(L - \lambda_1)\varphi_*\|_\infty$ by the maximum principle. However, this is too pessimistic. Since φ_* is a good approximation of φ_1 and $Q_t \varphi_1 = \varphi_1$, we expect $Q_t \varphi_* \approx \varphi_*$ as well. The key is that $(L - \lambda_1)\varphi_*$ is orthogonal to φ_1 . Therefore,

$$\begin{aligned} \|Q_s(L - \lambda_1)\varphi_*\|_2 &= e^{\lambda_1 s} \|P_s(L - \lambda_1)\varphi_*\|_2 \\ &\leq e^{\lambda_1 s} e^{-\lambda_2 s} \|(L - \lambda_1)\varphi_*\|_2 \\ &= e^{-(\lambda_2 - \lambda_1)s} \|(L - \lambda_1)\varphi_*\|_2. \end{aligned} \quad (46)$$

To get a bound in L^∞ we use Wang-Li-Yau, we obtain

$$\|Q_{s_1+s_2}(L - \lambda_1)\varphi_*\|_\infty \leq C_{s_1} \|Q_{s_2}(L - \lambda_1)\varphi_*\|_{L^2(e^{-V})} \leq C_{s_1} e^{-(\lambda_2 - \lambda_1)s_2} \|(L - \lambda_1)\varphi_*\|_{L^2(e^{-V})} \quad (47)$$

The problem here is that C_{s_1} has a singularity at $s_1 = 0$. A simple workaround is to use the naive maximum principle bound for small times and the above bound for later times, that is

$$\|\varphi_* - \varphi_1\|_\infty \leq \|(L - \lambda_1)\varphi_*\|_\infty \int_0^{s_1} e^{\lambda_1 s} ds + \|(L - \lambda_1)\varphi_*\|_2 \int_{s_1}^\infty C_{s/2} e^{-2(\lambda_2 - \lambda_1)s} ds. \quad (48)$$

We can get a very crude estimate by setting $s_1 = 3.0$. Then, the first integral evaluates to

$$\int_0^{s_1} e^{\lambda_1 s} ds = \frac{e^{3\lambda_1} - 1}{\lambda_1} \approx 20. \quad (49)$$

For the second integral we split $\frac{s}{2} = \frac{s-1}{2} + 0.5$ to obtain the constant

$$C_{s/2} \leq \frac{\sqrt{M}}{3} \frac{3}{2} = \frac{\sqrt{M}}{2} \approx 2 \quad (50)$$

As a result, the second interval evaluates to

$$\int_3^\infty C_{s/2} e^{-2(\lambda_2 - \lambda_1)s} ds \leq 2 \int_3^\infty e^{-2(\lambda_2 - \lambda_1)s} ds = \frac{e^{-6(\lambda_2 - \lambda_1)}}{\lambda_2 - \lambda_1} \approx 0.0025, \quad (51)$$

assuming $\lambda_2 - \lambda_1 \approx 1$.

I left a lot of slack everywhere, so we can do much better, in particular for the first constant. We should be able to get away with a significantly smaller s_1 .

5.2.2 Barriers for the finite dimensional case

Suppose φ_* is an approximation of φ_1 in the sense that

$$\begin{aligned} -\Delta \varphi_* &= \lambda_* \varphi_* \\ \varphi_* &= 0 \text{ on } \Gamma_0 \\ \partial_{\vec{n}} \varphi_* &= 0 \text{ on } \Gamma_1 \\ \|\partial_{\vec{n}} \varphi_*\|_\infty &\leq \varepsilon \text{ on } \Gamma_2. \end{aligned} \quad (52)$$

We cannot find barriers for φ directly, instead we bound the correction (see Moler-Payne) needed for φ_* satisfy the boundary condition. However then the corrected approximate eigenfunction is not a perfect eigenfunction in the interior any more. Moler-Payne show that this corrected eigenfunction is close to the true eigenfunction in the L^2 sense. We then upgrade this L^2 estimate to an L^∞ estimate using the Wang-Li-Yau inequality.

Let w be the correction we have to add to φ_* to satisfy the Neumann boundary condition, i.e.

$$\begin{aligned} -\Delta w &= 0 \\ w &= 0 \text{ on } \Gamma_0 \\ \partial_{\vec{n}} w(x, r) &= -\partial_{\vec{n}} \varphi_*(x, r) \text{ on } \Gamma_1 \cup \Gamma_2. \end{aligned} \tag{53}$$

Now $\varphi_* + w$ satisfies the boundary conditions exactly at the cost $-\Delta(\varphi_* + w) = \lambda \varphi_* \neq \lambda_*(\varphi_* + w)$. According to Moler-Payne it follows that

Lemma 5.2.1 ([11]).

$$\|\varphi_* - \varphi_1\|_2 \leq \frac{\|w\|_2}{\alpha} \left(1 + \frac{\|w\|_2^2}{\alpha^2} \right)^{\frac{1}{2}}, \tag{54}$$

where $\alpha = |\lambda_2 - \lambda_*|/\lambda_2 = 1 - \frac{\lambda_*}{\lambda_2} \leq 1 - \frac{\lambda_*}{\lambda_2}$.

$$\begin{aligned} |\lambda_* - \lambda_1| &\leq \frac{\sqrt{2}\|w\| + \|w\|^2}{1 - \|w\|} \lambda_* \\ \lambda_1 &\leq \frac{\lambda_*}{1 - \|w\|_2} \end{aligned} \tag{55}$$

Maybe the Rayleigh-quotient is a better upper bound of λ_1 :

$$\begin{aligned} \lambda_1 &\leq \int |\nabla \varphi_*|^2 = - \int \varphi_* \Delta \varphi_* + \int \varphi_* \partial_{\vec{n}} \varphi_* \\ &= \lambda_* + \int_{\Gamma_1} \varphi_* \partial_{\vec{n}} \varphi_* \end{aligned} \tag{56}$$

However, we want pointwise bounds and can now, since we transformed to an interior problem, use the same procedure as in the infinite dimensional case: $\|(-\Delta - \lambda_*)(\varphi_* + w)\|_\infty = \lambda_* \|w\|_\infty$, by the same argument as before we conclude that

Let's redo this.

$$\begin{aligned} &\|\varphi_* + w - \varphi_1\|_\infty \\ &\leq \|(-\Delta - \lambda_1)(\varphi_* + w)\|_\infty \int_0^{s_1} e^{\lambda_1 s} ds + \|(-\Delta - \lambda_1)(\varphi_* + w)\|_2 \int_{s_1}^\infty C_{s/2} e^{-2(\lambda_2 - \lambda_1)s} ds. \end{aligned} \tag{57}$$

We have

$$\begin{aligned} \|(-\Delta - \lambda_1)(\varphi_* + w)\| &\leq \|(-\Delta - \lambda_1)\varphi_*\| + \lambda_1 \|w\| \\ &\leq \varepsilon + |\lambda_* - \lambda_1| \|\varphi_*\| + \lambda_1 \|w\|, \end{aligned} \tag{58}$$

and $|\lambda_* - \lambda_1| \leq \frac{\sqrt{2}\|w\| + \|w\|^2}{1 - \|w\|} \lambda_*$.

$$\begin{aligned}
\|\varphi_* + w - \varphi_1\|_\infty &\leq e^{\lambda_1 t} (\lambda_* \|w\|_\infty + |\lambda_* - \lambda_1| \|\varphi_* + w\|_\infty + C_t \|\varphi_* + w - \varphi_1\|_2) \\
&\leq e^{\lambda_1 t} (\lambda_* \|w\|_\infty + |\lambda_* - \lambda_1| (\|\varphi_*\|_\infty + \|w\|_\infty) + C_t (\|\varphi_* - \varphi_1\|_2 + \|w\|_2))
\end{aligned} \tag{59}$$

In order to bound w we use a barrier estimate, let v be such that

$$\begin{aligned}
-\Delta v &\geq 0 \\
v &= 0 \text{ on } \Gamma_0 \\
\partial_{\vec{n}} v &\geq 0 \text{ on } \Gamma_1 \\
\partial_{\vec{n}} v(x, r) &\geq |\partial_{\vec{n}} \varphi_*| \text{ on } \Gamma_2.
\end{aligned} \tag{60}$$

Let $f := w - v$, then

$$\begin{aligned}
-\Delta f &\leq 0 \\
\partial_{\vec{n}} f &\leq 0
\end{aligned} \tag{61}$$

Now,

$$\|\nabla f^+\|_2^2 = \int_{\Omega} \nabla f \cdot \nabla f^+ = \int_{\Omega} (-\Delta f) f^+ + \int_{\Gamma_1 + \Gamma_2} (\partial_{\vec{n}} f) f^+ \leq 0, \tag{62}$$

showing f^+ is constant. Since f^+ vanishes on Γ_0 it must be zero everywhere, we conclude that $w \leq v$. The same argument with $-w$ in place of w gives $|w| \leq v$.

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A Computations

A.1 Radial part of the MPS basis

Let $f(z) = z^{-\alpha} I_{\alpha}(z)$

$$\begin{aligned} f'(z) &= -\alpha z^{-\alpha-1} I_{\alpha}(z) + z^{-\alpha} I'_{\alpha}(z) \\ f''(z) &= \alpha(\alpha+1) z^{-\alpha-2} I_{\alpha}(z) - 2\alpha z^{-\alpha-1} I'_{\alpha}(z) + z^{-\alpha} I''_{\alpha}(z) \end{aligned} \quad (63)$$

$$\begin{aligned} f''(z) + \frac{2\alpha+1}{z} f'(z) &= z^{-\alpha-2} (z^2 I''_{\alpha}(z) + z I'_{\alpha}(z) - \alpha^2 I_{\alpha}(z)) \\ &= z^{-\alpha-2} (z^2 I_{\alpha}(z)) \\ &= z^{-\alpha} I_{\alpha}(z) \\ &= f(z) \end{aligned} \quad (64)$$

For $\alpha = \frac{d-1}{2}$ this simplifies to

$$f''(z) + \frac{d}{z} f'(z) = f(z). \quad (65)$$

Now substitute $z = \beta r$

$$\begin{aligned} \left(\frac{4}{d} \partial_r^2 + \frac{4}{r} \partial_r \right) f(\beta r) &= \frac{4}{d} \beta^2 f''(\beta r) + \frac{4}{r} \beta f'(\beta r) \\ &= \frac{4}{d} \beta^2 f''(\beta r) + \frac{4}{\beta r} \beta^2 f'(\beta r) \\ &= \frac{4}{d} \beta^2 \left(f''(\beta r) + \frac{d}{\beta r} f'(\beta r) \right) \\ &= \frac{4}{d} \beta^2 f(\beta r) \\ &= \frac{4}{d} \sqrt{\frac{d}{4} |\lambda|}^2 f(\beta r) \\ &= -\lambda f(\beta r) \end{aligned} \quad (66)$$

since $\lambda < 0$.

For the branch $\lambda \geq 0$ the proof is analogous with $z^2 J''_{\alpha}(z) + z J'_{\alpha}(z) + (z^2 - \alpha^2) I_{\alpha}(z) = 0$ instead.

A.2 Wang-Li-Yau Constants

A.2.1 $d = \infty$

The first constant. We want to find $C_1(t_1)$ such that

$$\int \exp\left(-\frac{|x|^2}{t_1} - \varepsilon V(x)\right) dx \geq \frac{1}{C_1}. \quad (67)$$

In the core $(-\frac{\pi}{2}, \frac{\pi}{2}) \times (-1, 1)$ the potential is approximately $\frac{M}{2}\|x\|^2$. Let $A := t_1^{-1} + \varepsilon M$, then

$$\begin{aligned} \int_{(-\frac{\pi}{2}, \frac{\pi}{2}) \times (-1, 1)} \exp(-A|x|^2) dx &= \frac{\pi}{A} \operatorname{erf}\left(\frac{\pi}{2}\sqrt{A}\right) \operatorname{erf}(\sqrt{A}) \\ &\geq \frac{\pi}{A} \operatorname{erf}(\sqrt{M})^2 \end{aligned} \quad (68)$$

For $M \approx 15$ and $t_1 \approx 1$ we obtain

$$\approx \frac{\pi}{16} \approx \frac{1}{5}. \quad (69)$$

Hence, $C_1 \approx 5$.

The second constant. The goal is to find C_2 such that

$$\left\| P_{t_2} \exp\left(\frac{\|x\|^2}{4t_1}\right) \right\|_{\infty} \leq C_2. \quad (70)$$

This one we estimate with a barrier, i.e. we want to find $G(t, x)$ such that

$$\begin{aligned} \partial_t G(t, x) + LG(t, x) &\geq 0 \\ G(0, x) &\geq e^{\alpha\|x\|^2}. \end{aligned} \quad (71)$$

Then we have $G(t, x) \geq P_t e^{\alpha\|x\|^2}$.

First we separate out x_2 , i.e. $P_t e^{\alpha\|x\|^2} \leq e^{\alpha x_2^2} P_t e^{\alpha x_1^2}$ making the problem effectively one dimensional. We now try to find a supersolution $G(t, x_1)$, i.e.

$$\begin{aligned} -L_1 G(t, x_1) &\leq \partial_t G(t, x_1) \\ G(0, x_1) &\geq e^{\alpha x_1^2}, \end{aligned} \quad (72)$$

where $L_1 = -\partial_1^2 + (\partial_1 V)\partial_1$. We try the ansatz $G(t, x_1) = \exp(a(t)x_1^2 + b(t))$, with $a(0) = \alpha$ and $b(0) = 0$.

$$\begin{aligned} \partial_t G(t, x_1) &= (a'(t)x_1^2 + b'(t))G(t, x_1) \\ \partial_1 G(t, x_1) &= 2a(t)x_1 G(t, x_1) \\ -L_1 G(t, x) &= (2a(t) + 4a(t)^2 x_1^2 - 2a(t)x_1 \partial_1 V)G(t, x_1) \end{aligned} \quad (73)$$

To get an upper bound of $-L_1 G$ we lower bound $x_1 \partial_1 V$. In the core we have $x_1 \partial_1 V \approx \varepsilon M x_1^2$. In the wings $\partial_1 V$ is huge, but it does depend on x_2 , however we claim that is much bigger than $\varepsilon M x_1^2$ everywhere. We obtain the bound

$$-L_1 G(t, x) \leq ((4a(t)^2 - 2\varepsilon M a(t))x_1^2 + 2a(t))G(t, x_1). \quad (74)$$

If we find a such that $a'(t) = 4a(t)^2 - 2\varepsilon M a(t)$ and b such that $b'(t) = 2a(t)$, then

$$-L_1 G(t, x_1) \leq (a'(t)x_1^2 + b'(t))G(t, x_1) = \partial_t G(t, x_1) \quad (75)$$

and we win. The following functions should to the trick:

$$\begin{aligned}
a(t) &= \frac{\varepsilon M}{2 + \left(\frac{\varepsilon M}{\alpha} - 2\right)e^{2\varepsilon M t}} \\
b(t) &= \varepsilon M t + \frac{1}{2} \ln\left(\frac{a(t)}{\alpha}\right)
\end{aligned} \tag{76}$$

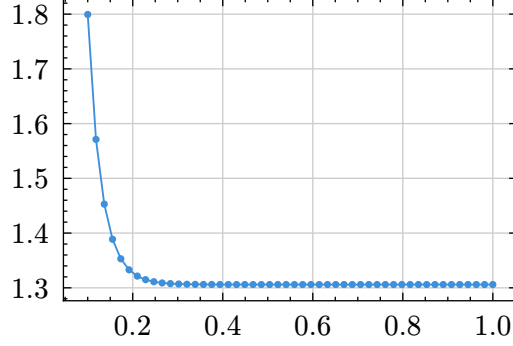


Figure 5: $C_2(t_2)$ evaluated at $t_1 = 1$ for $M = 15$ and $\max x_1 = 5$

A better estimate of the second constant at small times.

We have

$$P_{t_2} \exp\left(\frac{\|x\|^2}{4t_1}\right) = \mathbb{E}\left[\exp\left(\frac{\|X_t\|^2}{4t_1}\right) \mid X_0 = x\right] \tag{77}$$

where $dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t$. In the wings ∇V dominates, therefore, $dX_t \approx -\nabla V(X_t) dt$. Assume for now that $\partial_y V = 0$ in the wings, then

$$X_t \approx X_0 - t \nabla V. \tag{78}$$

Assuming the wing has length l_{wing} , then X_t reaches the core boundary at $t = (l_{\text{wing}} - X_0)/\nabla V$. Since $\nabla V \approx -\varepsilon 10^7$ this time is very small. Now do the analysis from the semester paper. This should lead a constant that is approximately the infinity norm in the core.

A.2.2 $d < \infty$